



Analytical modeling of resonant converters in frequency domain for wireless power transfer: Continuous current mode (CCM) operation

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ARTICLE INFO

Article history:

Received 29 May 2021

Received in revised form

23 December 2021

Accepted 31 January 2022

Available online 3 February 2022

Keywords:

Wireless power transfer

Continuous current mode (CCM)

Analytical modelling

Frequency domain

ABSTRACT

The major topology for wireless power transfer is resonant converters, which use fundamental harmonic approximation as a general analysis method. The accuracy of fundamental harmonic approximation (FHA) is unsatisfactory in some wireless power transfer applications, as the current has distortions due to the non-linear diode-bridge rectifier. Some frequency-domain models also consider harmonics beside fundamental component and indeed have a higher resolution. However, due to the difficulty of modeling the non-linear rectifier, most models are oversimplified by assuming a voltage source on the output to eliminate the rectifier. Moreover, the value of the voltage source is often treated as a fixed given number, which is not true for a resistive load. To cover a general case, this paper proposes a frequency-domain analytical model for resonant converters in continuous current mode (CCM) with the capability of considering any load and source situations in kHz frequency range. In terms of computation complexity, the proposed model only involves matrix operation (first-order). A simple algorithm has been chosen to demonstrate the fast convergency and high accuracy of the proposed model. The main findings are verified by the comparison between the theoretical calculation, time-domain simulation and experimental results, where the estimation of output power has 10% improvement than the FHA method.

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1. Introduction

Wireless charger has drawn extensive attention in the past few years due to its advantages of convenience, reliability, and environmental adaptation comparing with the conventional charging techniques [1]. Air gap between the coils, however, is one of the severe challenges in wireless charging. Resonant converters are the mainstream topologies to address the issue of air gap in wireless charging, which require accurate steady state modeling to estimate the power loss and optimize the power density. The steady state modeling is particularly a critical issue in the time domain since the computational time is highly depended on the time constant and initial values of the system, which is typically unpredictable.

Given the expensive costs of the calculation resources for this

task in the time domain, although it is suitable for transient analysis (e.g., state-space modeling [3], linearized dynamic modeling [4], rotating coordinate modelling [5], average geometric control modeling [6], etc.), frequency-domain modeling has attracted research attention and there are various steady state modeling methods for resonant converters in recent years, such as AC output power factor modeling [2], overall AC equivalent modeling [7], phase-modulated resonant power conversion techniques [8], and series resonant converter analysis [9,10]. Among them, fundamental harmonic approximation (FHA) is one of the most popular methods in wireless power transfer (WPT) as the power transferred is carried by the fundamental component in vast resonant topologies. However, due to the fast development of WPT, FHA exhibits its limitations. To reveal FHA's limitations in emerging WPT applications, a typical inductive wireless charger is shown in Fig. 1.

The inductive wireless charger consists of a H-bridge inverter, a resonant network and a diode-bridge rectifier. Due to the large air gap between the primary and the secondary sides, a series or parallel compensation capacitor is typically required on both the

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Nomenclature

Symbols Meaning

V_{in} (V)	DC Input voltage
v_H (V)	H-bridge output voltage
i_H (A)	H-bridge output current
i_1 (A)	Primary side current
i_2 (A)	Secondary side current
C_p (uF)	Primary side resonant capacitor
C_s (uF)	Secondary side resonant capacitor
L_M (uH)	Magnetic inductance
L_{kp} (uH)	Primary side leakage
L_{ks} (uH)	Secondary side leakage
N_1	Primary side turns
N_2	Secondary side turns
C_o (uF)	Output capacitor
L_o (uH)	Output filter
V_o (V)	Output voltage
R_L (Ω)	Resistive load
R_{ac} (Ω)	Equivalent AC resistor or RL
Z_o (Ω)	Output impedance
α (rad)	Offset angle
N	Dimension of the matrix
H	Maximum order of harmonic
ω (rad/s)	Angular fundamental frequency
v_{ac} (V)	Rectifier input voltage
i_{ac} (A)	Rectifier input current
S_v	Switching function of v_{ac}
S_i	Switching function of i_{ac}

$V_{ac,h}$ (V)	Amplitude of hth harmonic of v_{ac}
$\varphi_{ac,h}$ (rad)	Angle of hth harmonic of v_{ac}
v_{dc} (V)	Rectifier output voltage
i_{dc} (A)	Rectifier output current
$v_{dc,k}$ (V)	kth harmonic of v_{dc}
$\dot{I}_{dc,n}$ (A)	Phasor of nth harmonic of i_{dc}
$\dot{V}_{dc,n}$ (V)	Phasor of nth harmonic of v_{dc}
Z_n (Ω)	The nth order of output impedance to rectifier
θ_n (rad)	Angle of Z_n
I_{ac} (A)	A matrix form of i_{ac}
$I_{ac,n}$ (A)	The nth harmonic of I_{ac}
$Y_{N \times H}^+$ (Ω^{-1})	Admittance matrix of positive sequence
$Y_{N \times H}^-$ (Ω^{-1})	Admittance matrix of negative sequence
$Y_{n,h}^+$ (Ω^{-1})	The nth line and hth column element of $Y_{N \times H}^+$ matrix
$Y_{n,h}^-$ (Ω^{-1})	The nth line and hth column element of $Y_{N \times H}^-$ matrix
V_{ac} (V)	A matrix form of v_{ac}
$Z_{N \times H}^+$ (Ω)	Impedance matrix of positive sequence
$Z_{N \times H}^-$ (Ω)	Impedance matrix of negative sequence
$Z_{n,h}^+$ (Ω)	The nth line and hth column element of $Z_{N \times H}^+$ matrix
$Z_{n,h}^-$ (Ω)	The nth line and hth column element of $Z_{N \times H}^-$ matrix
V_H (V)	A matrix form of v_H
Z_{11}, Z_{12}, Z_{21} and Z_{22} (Ω)	Two-port network parameter matrices
Y_p (Ω^{-1})	A short form of $Y_{N \times H}^+$
Y_n (Ω^{-1})	A short form of $Y_{N \times H}^-$
I_E, I'_E and Y_E	Intermediate matrices
V_D (V)	Voltage drop of the diode
Y_o (Ω^{-1})	Admittance of R_o
$E_{N \times 1}$	Diodes' voltage drop compensation matrix

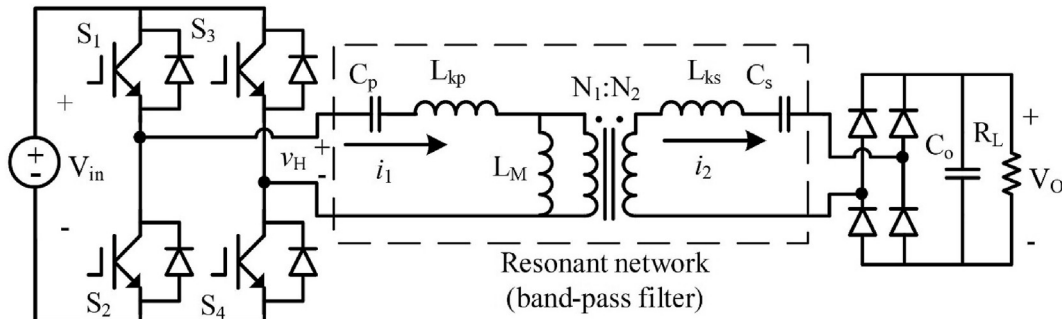


Fig. 1. Overall schematic diagram for the conventional inductive wireless charger.

primary and the secondary sides to cancel the leakage inductance. Note that the output rectifier together with the load can be modeled as an AC resistor by the FHA method [4–7], and the entire system can be simplified to a linear relationship for analysis. This FHA modeling method has also been adopted in other resonant converters [15,16]. Apart from such basic series or parallel compensation, an LCL network was proposed in Ref. [17], where the transmitter was featured as a current source. Aiming to reduce the size and weight of the additional inductor, an integrated LCC network was developed in Ref. [18]. In addition, a CLCL network for bidirectional wireless power transfer was presented in Ref. [19]. Numerous compensation networks have been developed to achieve different functions in WPT. When the operation point has a drift

away from the resonant point, the output current experiences certain distortion rather than maintains a pure sinusoidal waveform, which severely affects the accuracy of FHA. The third harmonic injection in resonant converters for higher power density has come to a thought where FHA is not applicable [20,27].

To achieve a higher accuracy model, higher order harmonics must be considered in the models by utilizing the harmonic matrix [8–10]. Accepting the non-linear behavior of diode-bridge rectifier, harmonic matrix modeling derives the current and voltage relationship between the AC side and the DC side, and the rest of the circuit remains linear. Researches, back in the 1980s, have derived the basic current and voltage relationship for diode-bridge rectifier [8]. When this model is applied to the system, most studies [9–11]

replace the diode-bridge with a square-wave voltage source, and thus, the load information is lost. The one presented in Ref. [12] requires a capacitive load after the rectifier, and a series inductor is needed in Ref. [13]. In particular, the harmonic matrix model in Ref. [13] is not strictly linear as the phase information of the fundamental component is required for the matrices. The study in Ref. [14] assumes an ideal sinusoidal supply voltage, which is difficult to be applied to the WPT system with the resonant network. Lack of effective harmonic matrix modeling method that can properly overcome the last barrier, diode-bridge rectifier, results in the dilemma that simplifications must be made to fulfill the modeling, e.g., ideal sinusoidal input or fixed output voltage source. Note that above modeling method only consider the non-linear characteristic caused by the switching behavior of rectifier, but not considering junction capacitor, other rectifier parasitic components or the non-linear magnetic material loss.

From the mathematical aspect, no exact analytical matrix model has been reported in the existing studies to describe the resonant converters without any assumptions or any unknown variables. Table 1 reviews the harmonic matrix modeling in the existing studies. Although the voltage and current relationship of diode-bridge is presented in Ref. [8], the full system analysis is still limited to FHA. In particular, the total number of variables is $4N$ if the dimension of the matrix is N , up to $(2N-1)^{\text{th}}$ harmonic, including the harmonic voltages and currents from both rectifier input and output. The method in Ref. [8] eliminated N variables from the input and N from the output by inserting a low pass filter. The assumption of a given output erases N variables that represent the load [9–12]. The voltage source in Ref. [14] is the grid and therefore, N variables are neglected. Arbitrary input and inductive load are allowed with 2 unknowns in the equations reported in Ref. [13]. In general, the more the variables are considered, the larger the calculation cost is.

In this paper, a first order frequency-domain analytical model is proposed for resonant converters in CCM for WPT application in kHz range, such that to find the ultimate matrix expression accounts for arbitrary input and output. Comparing with the current status in the field, the key contributions are summarized as:

- The proposed model reduces the unknown variables to one. The assumption of specific input or output is not needed.
- The proposed method only involves matrix calculation without integration, differential or logic operation (i.e., first order terms). The unknown variable, offset angle, is decoupled with the matrix operation, while the ones in Ref. [13] are coupled, which further increases the speed of calculation.
- The proposed method allows the steady state analysis for the entire resonant converter in CCM without approximations for the first time. Thanks to the decoupled characteristic, the linear components in the resonant network can be combined to the matrix model of the non-linear rectifier. Note that $N = 50$

provides adequately accurate results from the engineering point of view.

The rest of the paper is organized as: **Section II** is the methodology which illustrates in three steps. First the modeling of the diode-bridge rectifier with load is provided. Then the resonant network is treated as a two-port network into the model. Lastly, the algorithms for searching the offset angle are explored. Section III compares the simulation, theoretical and experimental results using two WPT systems and satisfactory agreements are obtained. Section IV summarizes the main findings in this study. **Section II** have already been published in the previous research [26], the algorithm in Section II and experimental results in Section III are the new contributions to this topic.

2. Methodology

2.1. Modelling of diode-bridge rectifier

A WPT system can be simplified to three blocks, as shown in Fig. 2(a). The voltage source from a H-bridge inverter consists of all kinds of harmonics. The resonant network is usually a band-pass filter only for the fundamental frequency, and with isolated coils in WPT application. In this section, the modeling of the last part, diode-bridge rectifier, will be discussed. Fig. 2(b) presents the input waveforms of the diode-bridge rectifier. Three assumptions are taken into account in this modeling. The first assumption is that the rectifier runs in CCM such that the switching function S_i equals S_v . The second assumption is that the high-order currents are small enough comparing to the fundamental current and do not affect the shape of the switching function. The third assumption is that the operation frequency is low enough to ignore the parasitic components in a rectifier. The offset angle α is the zero-crossing point for i_{ac} with respect to the phase of the fundamental component of v_H . The switching function can be expressed in a sum of sequence as

$$S_v(t) = S_i(t) = \sum_{n=1,3,5,\dots}^{2N+1} \frac{2}{jn\pi} \left(e^{jn\omega t - jn\alpha} - e^{-jn\omega t + jn\alpha} \right), \quad (1)$$

where ω is the fundamental angular frequency of the system. The general expression of the input voltage v_{ac} is

$$v_{ac}(t) = \sum_{h=1,3,5,\dots}^{2H+1} \frac{\sqrt{2}V_{ac,h}}{2j} \left(e^{jh\omega t + j\varphi_{ac,h}} - e^{-jh\omega t - j\varphi_{ac,h}} \right), \quad (2)$$

To distinguish from the dimension of switching function, variable H is used instead of N . For h^{th} order harmonic, $V_{ac,h}$ is the amplitude and $\varphi_{ac,h}$ is the phase angle. The DC output voltage would be

Table 1
Review and comparison of the existing harmonic matrix modeling method and the proposed method.

Harmonic matrix modeling	Rectifier Input	Rectifier Output	Number of unknowns (equivalent)	Calculation cost	Demonstrated with the entire resonant converter
[8]	Fundamental	DC voltage	$2N$	Small	No
[9–11]	Arbitrary	Square wave	N	Medium	No
[12]	Arbitrary	Given voltage/current source	N	Medium	No
[14]	Sine wave	Arbitrary	N	Medium	No
[13]	Arbitrary	Inductive	2	Large	No
This study	Arbitrary	Arbitrary	1	Large	Yes

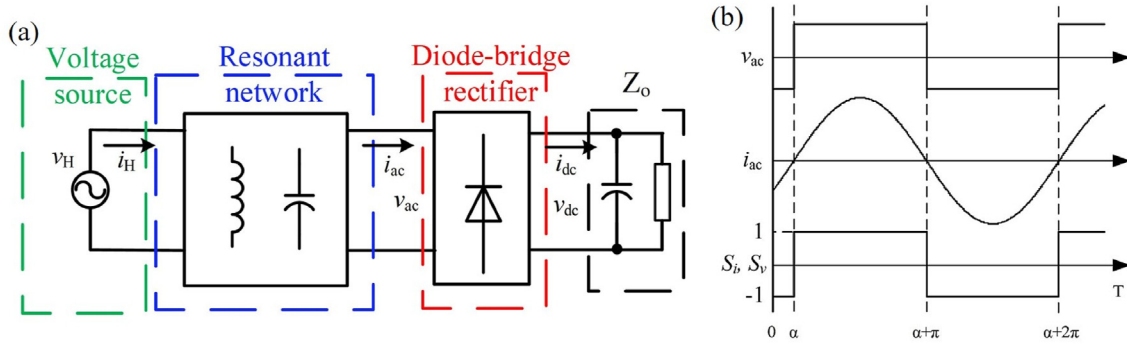


Fig. 2. (a) Simplifier circuit diagram of a WPT system. (b) Input waveforms of the diode-bridge rectifier.

$$v_{dc}(t) = S_v(t) \cdot v_{ac}(t) = V_{dc,0} + \sum_{k=2,4,6,\dots}^{2H+2N+2} v_{dc,k}(t), \quad (3)$$

where

$$v_{dc,k}(t) = \sum_{h=1,3,5,\dots}^{2H+1} \frac{\sqrt{2}V_{ac,h}}{(h-k)\pi} \left(e^{jk\omega t + j\varphi_{ac,h} + j(k-h)\alpha} + e^{-jk\omega t - j\varphi_{ac,h} - j(k-h)\alpha} \right) + \sum_{h=1,3,5,\dots}^{2H+1} \frac{\sqrt{2}V_{ac,h}}{(h+k)\pi} \left(e^{jk\omega t - j\varphi_{ac,h} + j(k+h)\alpha} + e^{-jk\omega t + j\varphi_{ac,h} - j(k+h)\alpha} \right) \quad (4)$$

Once \$v_{dc}\$'s phasor expression is obtained, \$i_{dc}\$ can be written as

$$I_{dc,n} = \dot{V}_{dc,n} / Z_n = \dot{V}_{dc,n} / (|Z_n| \angle \theta_n). \quad (5)$$

As illustrated in Fig. 3(a), \$Z_n\$ is the \$2n^{\text{th}}\$ order component of the output impedance \$Z_o\$ after the rectifier in the even orders. A simple LC filter is chosen as an example to demonstrate the impedance calculation. As there is a negative sequence in \$v_{dc}\$'s expression, the phase \$\theta_n\$ has an extra “-1” for the negative sequence when formula Eq. (5) is transferred back to the time domain. When the time-

In a matrix form, \$\mathbf{I}_{ac}\$ will be

$$\begin{bmatrix} I_{ac,1} \\ I_{ac,3} \\ \vdots \\ I_{ac,n} \end{bmatrix} = \begin{bmatrix} Y_{1,1}^+ & Y_{1,3}^+ & \cdots & Y_{1,h}^+ \\ Y_{3,1}^+ & Y_{3,3}^+ & \cdots & Y_{3,h}^+ \\ \vdots & \vdots & \ddots & \vdots \\ Y_{n,1}^+ & Y_{n,3}^+ & \cdots & Y_{n,h}^+ \end{bmatrix} \cdot \begin{bmatrix} V_{ac,1} \angle \varphi_{ac,1} \\ V_{ac,3} \angle \varphi_{ac,3} \\ \vdots \\ V_{ac,h} \angle \varphi_{ac,h} \end{bmatrix} + \begin{bmatrix} Y_{1,1}^- & Y_{1,3}^- & \cdots & Y_{1,h}^- \\ Y_{3,1}^- & Y_{3,3}^- & \cdots & Y_{3,h}^- \\ \vdots & \vdots & \ddots & \vdots \\ Y_{n,1}^- & Y_{n,3}^- & \cdots & Y_{n,h}^- \end{bmatrix} \cdot \begin{bmatrix} V_{ac,1} \angle -\varphi_{ac,1} \\ V_{ac,3} \angle -\varphi_{ac,3} \\ \vdots \\ V_{ac,h} \angle -\varphi_{ac,h} \end{bmatrix}$$

or

$$\mathbf{I}_{ac,N \times 1} = \mathbf{Y}_{N \times H}^+ \mathbf{V}_{ac,H \times 1} + \mathbf{Y}_{N \times H}^- \bar{\mathbf{V}}_{ac,H \times 1}, \quad (7)$$

where the elements in \$\bar{\mathbf{V}}_{ac}\$ are the conjugate of that in \$\mathbf{V}_{ac}\$. The detail of admittance matrix is

$$\begin{cases} Y_{n,h}^+ = \left[\frac{4}{hn\pi^2 Z_0} + \sum_{k=2,4,6,\dots}^{2N+2H+2} \frac{4}{(h-k)(n-k)\pi^2 Z_k} + \sum_{k=2,4,6,\dots}^{2N+2H+2} \frac{4}{(h+k)(n+k)\pi^2 \bar{Z}_k} \right] e^{-j(n-h)\alpha} \\ Y_{n,h}^- = \left[\frac{4}{hn\pi^2 Z_0} + \sum_{k=2,4,6,\dots}^{2N+2H+2} \frac{4}{(h-k)(n+k)\pi^2 \bar{Z}_k} + \sum_{k=2,4,6,\dots}^{2N+2H+2} \frac{4}{(h+k)(n-k)\pi^2 Z_k} \right] e^{-j(n+h)\alpha} \end{cases}. \quad (8)$$

domain \$i_{dc}\$ expression is available, the expression of \$i_{ac}\$ yields

$$i_{ac}(t) = S_i(t) \cdot i_{dc}(t) = \sum_{n=1,3,\dots}^{2N+1} \sum_{h=1,3,5,\dots}^{2H+1} \left(Y_{n,h}^+ \cdot V_{ac,h} \cos(n\omega t + \varphi_{ac,h}) + Y_{n,h}^- \cdot V_{ac,h} \cos(n\omega t - \varphi_{ac,h}) \right). \quad (6)$$

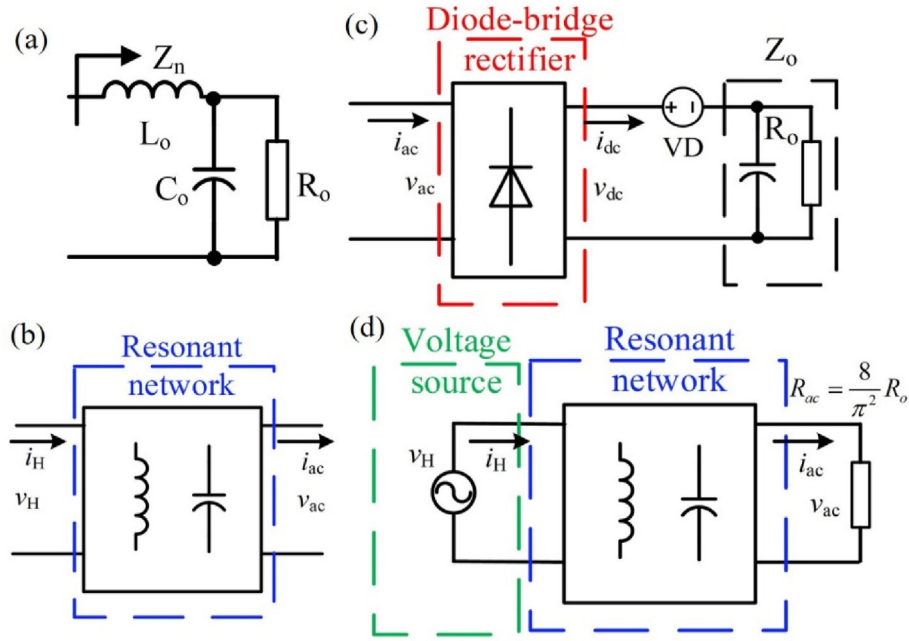


Fig. 3. (a) Output impedance Z_n . (b) Two port network model of resonant network. (c) Equivalent circuit for real diode-bridge rectifier. (d) Simplified FHA model for offset angle's initial value.

To construct the admittance matrix, the load information and offset angle are needed in Eq. (8). Note that the method in Ref. [13] still has the fundamental phase angle in every term of the admittance matrix. Offset angle can be solved in next section once the resonant network is determined.

As a dual method, the admittance matrix has an alternative expression, impedance matrix. The accuracy of the two methods will be discussed in the following sections once the final equation is derived. The only difference in the expressions is that Z_n is in the numerator instead of denominator as Eq. (10) shows, and the corresponding input relationship defined in Eq. (7) is changed to

$$\mathbf{V}_{ac, N \times 1} = \mathbf{Z}_{N \times H}^+ \mathbf{I}_{ac, H \times 1} + \mathbf{Z}_{N \times H}^- \mathbf{I}_{ac, H \times 1}, \quad (9)$$

and

2.2. Modelling of resonant converter

The resonant network in Fig. 3(b) only consists of the passive components which are all considered linear for simplified, although the experimental results show the magnetic material's non-linear loss characteristic affects the accuracy. Therefore, it can be represented as a two-port network with impedance matrix as below

$$\begin{cases} \mathbf{V}_H = \mathbf{Z}_{11} \mathbf{I}_H - \mathbf{Z}_{12} \mathbf{I}_{ac} \\ \mathbf{V}_{ac} = \mathbf{Z}_{21} \mathbf{I}_H - \mathbf{Z}_{22} \mathbf{I}_{ac} \end{cases} \quad (11)$$

It is worthwhile to point out that the variation of the coupling coefficient is reflected in the impedance matrix. These matrices only have diagonal elements because of the linear characteristic. The calculation of parameters in $\mathbf{Z}_{11}$, $\mathbf{Z}_{12}$, $\mathbf{Z}_{21}$ and $\mathbf{Z}_{22}$ are given as

$$\begin{cases} \mathbf{Z}_{n,h}^+ = \left[\frac{4Z_0}{hn\pi^2} + \sum_{k=2,4,6\dots}^{2N+2H+2} \frac{4Z_k}{(h-k)(n-k)\pi^2} + \sum_{k=2,4,6\dots}^{2N+2H+2} \frac{4\bar{Z}_k}{(h+k)(n+k)\pi^2} \right] e^{-j(n-h)\alpha} \\ \mathbf{Z}_{n,h}^- = \left[\frac{4Z_0}{hn\pi^2} + \sum_{k=2,4,6\dots}^{2N+2H+2} \frac{4\bar{Z}_k}{(h-k)(n+k)\pi^2} + \sum_{k=2,4,6\dots}^{2N+2H+2} \frac{4Z_k}{(h+k)(n-k)\pi^2} \right] e^{-j(n+h)\alpha} \end{cases} \quad (10)$$

$$\begin{cases} Z_{11}(n, n) = \frac{\mathbf{V}_H(n, n)|_{v_{ac} \text{ open}}}{\mathbf{I}_H(n, n)|_{v_{ac} \text{ open}}} \\ Z_{12}(n, n) = \frac{Z_{11}(n, n)\mathbf{I}_H(n, n)|_{v_H \text{ short}}}{\mathbf{I}_{ac}(n, n)|_{v_H \text{ short}}} \\ Z_{21}(n, n) = \frac{Z_{22}(n, n)\mathbf{I}_{ac}(n, n)|_{v_{ac} \text{ short}}}{\mathbf{I}_H(n, n)|_{v_{ac} \text{ short}}} \\ Z_{22}(n, n) = \frac{\mathbf{V}_{ac}(n, n)|_{v_H \text{ open}}}{-\mathbf{I}_{ac}(n, n)|_{v_H \text{ open}}} \end{cases} \quad (12)$$

2.2.1. Modeling with ideal rectifier

The voltage drop of the rectifier is zero in this ideal model. To combine with the rectifier model, the conjugate term of $\mathbf{V}_{ac}$ is eliminated. Taking conjugate on both side of Eq. (7) and eliminating the conjugate term with original Eq. (7), $\mathbf{V}_{ac}$ is obtained as

$$\mathbf{V}_{ac} = \left(\bar{\mathbf{Y}}_n - \bar{\mathbf{Y}}_p \mathbf{Y}_n^{-1} \mathbf{Y}_p \right)^{-1} \bar{\mathbf{I}}_{ac} - \left(\bar{\mathbf{Y}}_n - \bar{\mathbf{Y}}_p \mathbf{Y}_n^{-1} \mathbf{Y}_p \right)^{-1} \bar{\mathbf{Y}}_p \mathbf{Y}_n^{-1} \mathbf{I}_{ac}, \quad (13)$$

where $\mathbf{Y}_p$ stand for $\mathbf{Y}^+$ and $\mathbf{Y}_n$ stand for $\mathbf{Y}^-$. In Eq. (13), the $\mathbf{Y}_n$ matrix may not be full rank and the inverted $\mathbf{Y}_n$ does not exist. $\mathbf{Y}_p$ is invertible as there is a dominant non-zero term on diagonal elements, and therefore, an alternative solution is to eliminate $\mathbf{V}_{ac}$ in Eq. (7) as

$$\bar{\mathbf{V}}_{ac} = \left(\bar{\mathbf{Y}}_p - \bar{\mathbf{Y}}_n \mathbf{Y}_p^{-1} \mathbf{Y}_n \right)^{-1} \bar{\mathbf{I}}_{ac} - \left(\bar{\mathbf{Y}}_p - \bar{\mathbf{Y}}_n \mathbf{Y}_p^{-1} \mathbf{Y}_n \right)^{-1} \bar{\mathbf{Y}}_n \mathbf{Y}_p^{-1} \mathbf{I}_{ac}. \quad (14)$$

Eliminating $\mathbf{I}_H$ from Eq. (11), $\mathbf{I}_{ac}$ can be expressed as

$$\mathbf{I}_{ac} = \left(\mathbf{Z}_{22} - \mathbf{Z}_{21} \mathbf{Z}_{11}^{-1} \mathbf{Z}_{12} \right)^{-1} \cdot \left(\mathbf{Z}_{21} \mathbf{Z}_{11}^{-1} \mathbf{V}_H - \mathbf{V}_{ac} \right). \quad (15)$$

Inserting Eq. (14) into Eq. (7), a similar form as Eq. (7) is obtained as

$$\mathbf{I}_E = \mathbf{Y}_E \mathbf{V}_{ac} + \mathbf{Y}_n \bar{\mathbf{V}}_{ac}, \quad (16)$$

where $\mathbf{I}_E = \left(\mathbf{Z}_{22} - \mathbf{Z}_{21} \mathbf{Z}_{11}^{-1} \mathbf{Z}_{12} \right)^{-1} \cdot \mathbf{Z}_{21} \mathbf{Z}_{11}^{-1} \mathbf{V}_H$ and $\mathbf{Y}_E = \mathbf{Y}_p + \left(\mathbf{Z}_{22} - \mathbf{Z}_{21} \mathbf{Z}_{11}^{-1} \mathbf{Z}_{12} \right)^{-1}$.

As $\mathbf{Y}_E$ contains the information of the impedance matrix, $\mathbf{Y}_E$ is full rank as $\mathbf{Z}_{22} - \mathbf{Z}_{21} \mathbf{Z}_{11}^{-1} \mathbf{Z}_{12}$ is full rank. Therefore, instead of Eq. (13), applying Eq. (14) to Eq. (16), the final analytical expression is

$$\bar{\mathbf{V}}_{ac} = \left(\bar{\mathbf{Y}}_E - \bar{\mathbf{Y}}_n \mathbf{Y}_E^{-1} \mathbf{Y}_n \right)^{-1} \bar{\mathbf{I}}_E - \left(\bar{\mathbf{Y}}_E - \bar{\mathbf{Y}}_n \mathbf{Y}_E^{-1} \mathbf{Y}_n \right)^{-1} \bar{\mathbf{Y}}_n \mathbf{Y}_E^{-1} \mathbf{I}_E. \quad (17)$$

In Eq. (17), $\mathbf{V}_H$ is the excitation voltage and the $\mathbf{Y}$ matrices contain both the wireless resonant network and the load information. The only unknown variable is the offset angle α . The offset angle can be found by iteration and the constraint equation is that $i_{ac}(t)$ should be zero at angle α . For impedance matrix, the dual equations of Eq. (17) would be $\bar{\mathbf{I}}_{ac}$. Since $\bar{\mathbf{I}}_{ac}$ usually has less high order harmonics, impedance matrix would have slightly higher accuracy in the case of a finite N and H . In summary, the algorithm of solving the model with ideal rectifier is given in below:

- 1) Based on the load, choose admittance or impedance matrix form.
- 2) Pick up an arbitrary offset angle α .
- 3) Calculate Eq. (17), and use Eq. (7) to obtain $i_{ac}(t)$.
- 4) Choose a new α and repeat step 3 if $i_{ac}(t_\alpha)$ is not closed to zero.

2.2.2. Modeling with diodes' voltage drop

For high output voltage, the rectifier's voltage drop can be negligible. However, for low output voltage or high frequency application, the voltage drop of the rectifier contributes a considerable error to the results. This section serves as a patch for the ideal model. Fig. 3(c) demonstrates that the voltage drop $2V_D$ can be modeled as a DC voltage source between the DC side of the rectifier and the load, while the body of the rectifier remains its ideal characteristic. Therefore, most of the equations derived above could still hold true. The DC side voltage $v_{dc}(t)$ has the term of $2V_D$ in Eq. (3), which leads to one more term in Eq. (7) as

$$\begin{aligned} \mathbf{I}_{ac, N \times 1} &= \mathbf{Y}_{N \times H}^+ \mathbf{V}_{ac, H \times 1} + \mathbf{Y}_{N \times H}^- \bar{\mathbf{V}}_{ac, H \times 1} - \mathbf{S}_{i, N \times 1} \frac{2V_D}{R_o} \\ &= \mathbf{Y}_{N \times H}^+ \mathbf{V}_{ac, H \times 1} + \mathbf{Y}_{N \times H}^- \bar{\mathbf{V}}_{ac, H \times 1} - \mathbf{Y}_o \mathbf{E}_{N \times 1}. \end{aligned} \quad (18)$$

where $\mathbf{S}_i$ is the matrix form of switching function $S_i(t)$. The following symbols are defined for $\mathbf{Y}_o$ and $\mathbf{E}$

$$\mathbf{Y}_o = \frac{1}{R_o} \text{ and } \mathbf{E}_{n,1} = \frac{8V_D}{n\pi} \angle -n\alpha. \quad (19)$$

Inserting Eq. (15) into Eq. (18), an equation similar to Eq. (7) is obtained as

$$\mathbf{I}'_E = \mathbf{Y}_E \mathbf{V}_{ac} + \mathbf{Y}_n \bar{\mathbf{V}}_{ac}, \quad (20)$$

where $\mathbf{I}'_E = \left(\mathbf{Z}_{22} - \mathbf{Z}_{21} \mathbf{Z}_{11}^{-1} \mathbf{Z}_{12} \right)^{-1} \times \mathbf{Z}_{21} \mathbf{Z}_{11}^{-1} \mathbf{V}_H + \mathbf{Y}_o \mathbf{E}$ and $\mathbf{Y}_E = \mathbf{Y}_p + \left(\mathbf{Z}_{22} - \mathbf{Z}_{21} \mathbf{Z}_{11}^{-1} \mathbf{Z}_{12} \right)^{-1}$. Therefore, applying Eq. (14) to Eq. (20), the final analytical expression is

$$\bar{\mathbf{V}}_{ac} = \left(\bar{\mathbf{Y}}_E - \bar{\mathbf{Y}}_n \mathbf{Y}_E^{-1} \mathbf{Y}_n \right)^{-1} \bar{\mathbf{I}}'_E - \left(\bar{\mathbf{Y}}_E - \bar{\mathbf{Y}}_n \mathbf{Y}_E^{-1} \mathbf{Y}_n \right)^{-1} \bar{\mathbf{Y}}_n \mathbf{Y}_E^{-1} \mathbf{I}'_E. \quad (21)$$

In summary, the diode voltage dropping results in a small DC offset in the variable $\mathbf{I}_E$ and the corresponding $\mathbf{V}_{ac}$, but the major steps in this modeling remain the same as modeling with ideal rectifier.

2.3. Algorithm of solving offset angle

As illustrated in Section III, the frequency-domain solution can be verified in the time domain. Instead of judging one point at offset angle, the sum of the difference between AC side and DC side is chosen as the objective function. This mathematical problem can be written as a single variable optimization:

$$\begin{cases} \text{Minimize } f(\alpha) = \sum_0^{2\pi} ||i_{ac}(\alpha) - i_{dc}(\alpha)||, \\ \text{Subjected to } 0 \leq \alpha < 2\pi \end{cases} \quad (22)$$

where $f(\alpha)$ calculates the error sum of the full range. Proposing an estimated initial value for the offset angle, Eq. (22) is considered as the minimization of a unimodal function around the initial value. The parabolic interpolation algorithm [21], i.e., the one-dimensional direct search method (DSM), is used to address the

unimodal minimization with high efficiency of convergence.

2.3.1. Determination of the initial value

Although FHA is not accurate for diode-bridge rectifier, it gives a close offset angle and it is linear. As demonstrated in Fig. 3(d), the non-linear component is removed, and either the DC resistor or the AC resistor is directly parallel with the two-port resonant network. Choosing v_H as the reference, the estimated phase of i_{ac} is expressed as

$$\alpha_0 = \angle \left(\frac{\mathbf{V}_H(1, 1)}{\mathbf{Z}_{11}(1, 1)(\mathbf{Z}_{22}(1, 1) + R_{ac}) - \mathbf{Z}_{12}(1, 1)} \right). \quad (23)$$

2.3.2. DSM algorithm

Using the predefined initial value α_0 , Eq. (22) can be summarized as how to find the global minimum of the unimodal function. Note that a series of one-dimensional DSMs are available, which are able to promptly obtain the global minimum using a properly pre-derived initial value. In general, the DSMs consist of two calculation steps:

- 1) Finding a unimodal interval in certain range where the minimum exists; and
- 2) Choosing a new value for the variable to reduce the range.

In particular, the first step is implemented by finding three points as: $\alpha_1 < \alpha_2 < \alpha_3$ and $f(\alpha_1) > f(\alpha_2) < f(\alpha_3)$. Since α_1 and α_3 are adjacent to α_2 with the least function value in the current step, a minimum can be obtained between α_1 and α_3 . The second step can be conducted by different approaches such as the Golden section search and Fibonacci search [22]. Here, the parabolic interpolation method is used. The new value of the variable is determined using the Cramer's rule [23]:

$$\alpha_4 = \frac{1}{2} \frac{(\alpha_2^2 - \alpha_3^2)f(\alpha_1) + (\alpha_3^2 - \alpha_1^2)f(\alpha_2) + (\alpha_1^2 - \alpha_2^2)f(\alpha_3)}{(\alpha_2 - \alpha_3)f(\alpha_1) + (\alpha_3 - \alpha_1)f(\alpha_2) + (\alpha_1 - \alpha_2)f(\alpha_3)}. \quad (24)$$

Fig. 4 indicates the flowchart of DSMs. According to the two steps of the DSMs, the flowchart is separated into two parts by the dotted boxes. Following the calculation steps, the minimization problem in Eq. (22) can be solved.

2.3.3. Performance of the algorithm

Other algorithms can be used to solve the minimization problem such as the enumeration [24] or the heuristic strategies, including the genetic algorithm and simulated annealing etc. [25]. However, relatively high computational cost (i.e., large numbers of iterations) is required for them to converge and hence, the efficiency is low when sweeping full range with high accuracy. Fig. 5(a) and (b) show the two example cases of solving Eq. (22) using the enumeration method. The local minima exist next to the global minimum and, more importantly, the function is more likely nondifferentiable, which leads to the un-solvability of the Newton method and gradient descend. Fig. 5(c) shows the performance of enumeration on solving Eq. (22). Furthermore, the algorithms are compared in terms of the iteration numbers between the optimal solutions, as shown in Fig. 5(d). The number of iterated values of α is taken as the metric of efficiency.

According to the findings, the parabolic interpolation algorithm is more efficient than the others. Although the enumeration can be used to solve the global optimization, the computational cost is high for full range sweep. It is worth pointing out that the Newton method might have more rapid convergence in theoretical analysis; however, it is insufficient in this problem since the function is nondifferentiable. In summary, the parabolic interpolation algorithm with the estimated initial value can achieve the minimum within ten iterations.

2.4. Computational complexity comparison

In the proposed algorithm, the number of iterations is small (i.e., usually less than 10). The most time-consuming steps are the

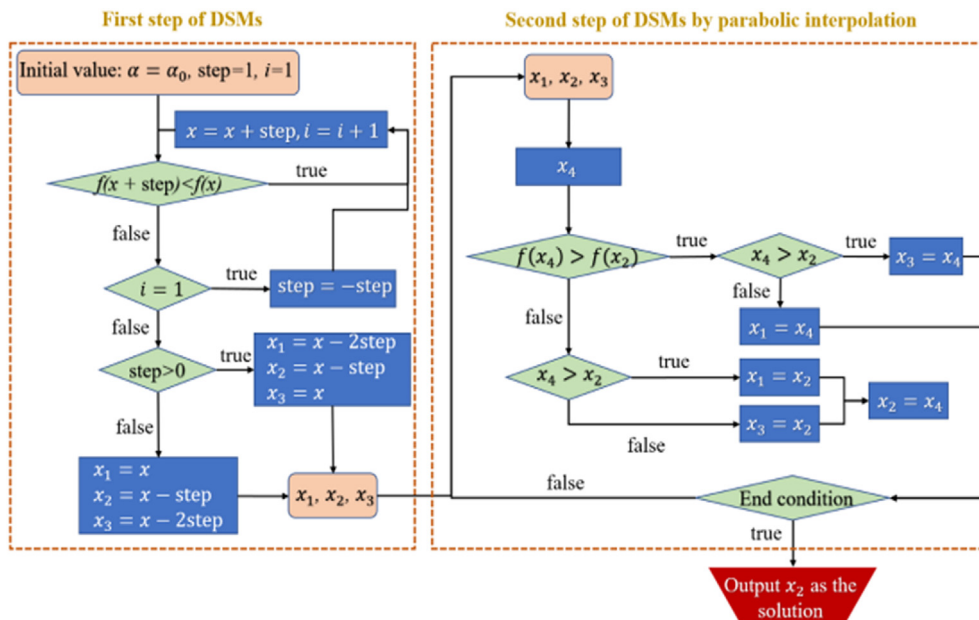


Fig. 4. Flowchart of DSMs.

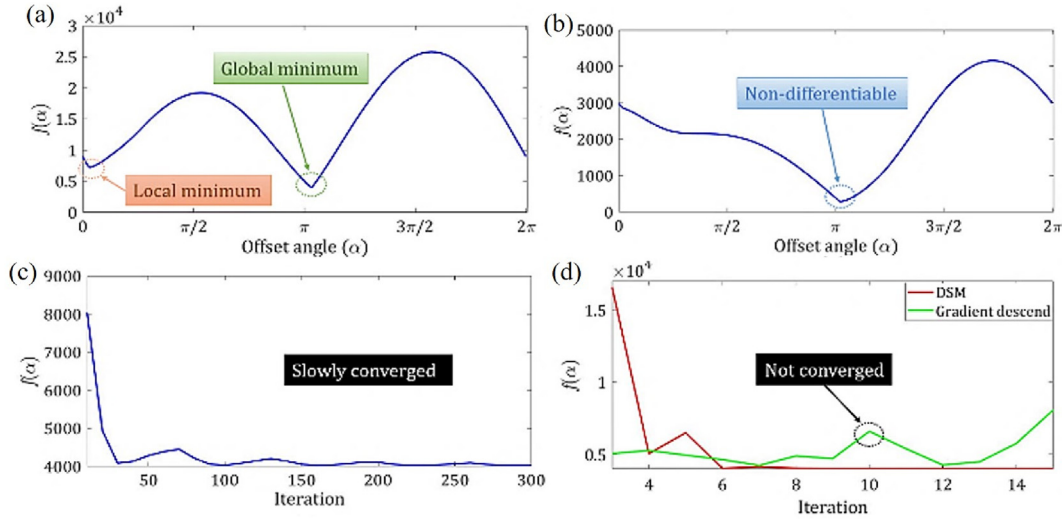


Fig. 5. Performance of different algorithms. (a) SS compensation full range sweep (b) LCC compensation full range sweep (c) Performance of Enumeration (d) Comparison between gradient descent and DSM.

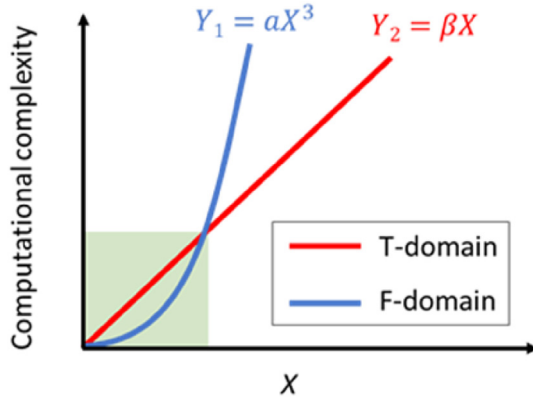


Fig. 6. Computational complexity for the proposed algorithm and time-domain simulation.

multiplication and inversion of a matrix. In general, the computational complexity of a $n \times n$ matrix multiplication and inversion is $O(n^3)$ [28,29]. The computational complexity of the proposed algorithm can be written as $Y_1 = \alpha X^3$, where the scale factor X is n . When X increases, the accuracy increases. For time-domain simulation, the number of variables depends on the number of components and is relatively small. However, it needs a huge number of iterations to reach steady state. For a fixed length of time T_0 , the accuracy is controlled by time step T , and the number of iterations is T_0/T . The computational complexity of time-domain simulation can be written as $Y_2 = \beta X$, where the scale factor X is $1/T$. By defining the scale factor X , the computational complexity of the proposed algorithm in the frequency domain and that of time-domain simulation can be illustrated in Fig. 6 for comparison. The proposed algorithm has advantage over time-domain simulation at the low X range. As X increases, the computational complexity for the proposed method increases dramatically, while the time-domain simulation maintains a linear relationship with X .

3. Results and discussion

To test the proposed frequency-domain model, two cases are demonstrated. An SS compensated wireless system for low power

application is built as shown in Fig. 7(a). An LCC compensated wireless system is modified from SS topology by inserting two LC filters as shown in Fig. 7(b).

Note that this LCC system is not fully optimized and the purpose is to show the efficacy of the proposed model in a more complex resonant network and a harmonic rich environment. The detail parameters are summarized in Table 2. In this section, the speed and accuracy of the algorithm are first discussed. Then, the proposed method is compared with conventional FHA modeling method, time-domain simulation and prototype experimental results with the same parameters in Table 2.

3.1. Algorithm performance

In general, the proposed model is only for CCM operation, since the light load DCM operation leads to different switching functions. Fig. 8 shows the error ($f(\alpha)$) variation when R_o increases. Observing from the time-domain simulation, the discontinuous boundary is approximately at 35 Ω for SS topology and 15 Ω for LCC topology. The corresponding error increases dramatically, which indicates the insufficiency of the proposed model in DCM.

Two typical cases are listed in Fig. 9 in terms of the time cost and accuracy. When the SS system works in a deeper CCM region with $R_o = 15 \Omega$, the error function is lower than the case closed to the DCM boundary ($R_o = 30 \Omega$). However, larger N in both cases significantly improves the accuracy at the cost of time. $N = 100$ consumes over 30 times of calculation resource than that of $N = 30$. The knee point of the error function is around $N = 30$, and therefore, $N > 30$ is obtained as an acceptable number for the proposed modeling and algorithm. Note that even in CCM, the error increases when the load is reduced, and decreases when N increases. The truncation of the matrix to N dimension leads to the error at light load conditions. N needs adjustment when approaching the boundary.

3.2. Experimental verification

The experimental setup is a dual coil transmitter designed for SS topology, originally for Qi standard receiver's wireless charging. To verify the proposed model, only one channel of the transmitter is used and modified as LCC topology as a second example to demonstrate the efficacy of the proposed modeling method. The

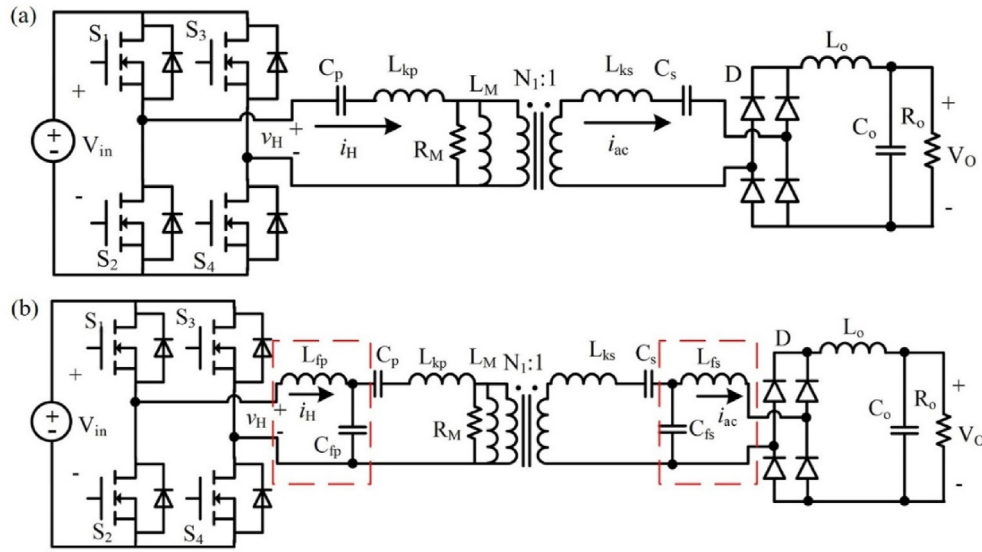


Fig. 7. Schematic of the test systems, (a)SS topology, (b)LCC topology.

Table 2
Test system parameters and values.

Parameter	Value	Parameter	Value
V_{in}	around 12 V	N_1	1.048
MOSFET	DMG7430LFG ($R_{ds,on} = 15 \text{ m}\Omega$)	D	DFLS130L ($V_D = 0.3\text{V}$)
L_{kp}	3.56 μH	ESR of L_{kp}	13 m Ω
L_{ks}	3.26 μH	ESR of L_{ks}	252 m Ω
L_o	0.56 μH	ESR of L_o	39 m Ω
L_M	12.51 μH	R_M	498 Ω
L_{fp}	9.65 μH	ESR of L_{fp}	0.19 Ω
L_{fs}	9.9 μH	ESR of L_{fs}	0.16 Ω
C_p	247 nF	C_s	400 nF
C_{fp}	149 nF	C_{fs}	10 nF
C_o	15 μF	R_o	10 Ω /20 Ω
Fundamental frequency	160 kHz	H-bridge dead time	250 ns

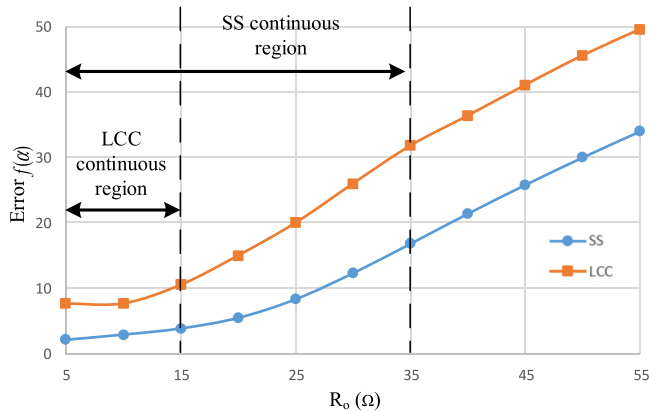


Fig. 8. Error vs. R_o for SS and LCC topologies ($V_{in} = 12 \text{ V}$, $N = 50$).

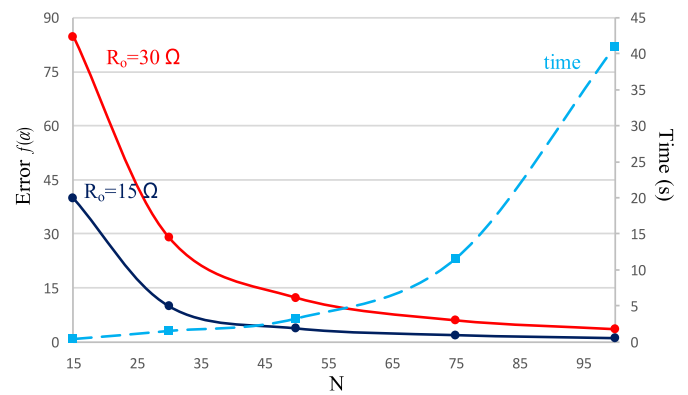


Fig. 9. Error vs. N and time cost for SS topology ($V_{in} = 12 \text{ V}$).

prototype photo is presented in Fig. 13. The prototype is operating in the open loop condition with the two coils fixed by tape. The coupling parameters are tested by LCR meter after power down. All parameters are recorded in Table 2. Note that the compensation capacitor's material is COG, where the capacitance is almost a constant with different DC bias.

Fig. 10 shows the comparison of the time-domain simulation,

calculation of the proposed method, calculation of FHA method and experimental results for SS topology. The time-domain simulation is conducted in commercial software PSIM. In FHA method, the rectifier is replaced by an AC resistor as Eq. (23) shows, and the entire system becomes linear with the fundamental frequency only. The results are aligned into one graph based on primary side switching waveforms in Fig. 10(b). Even though there is 250 ns deadtime from the controller, the prototype in ZVS mode leads to a

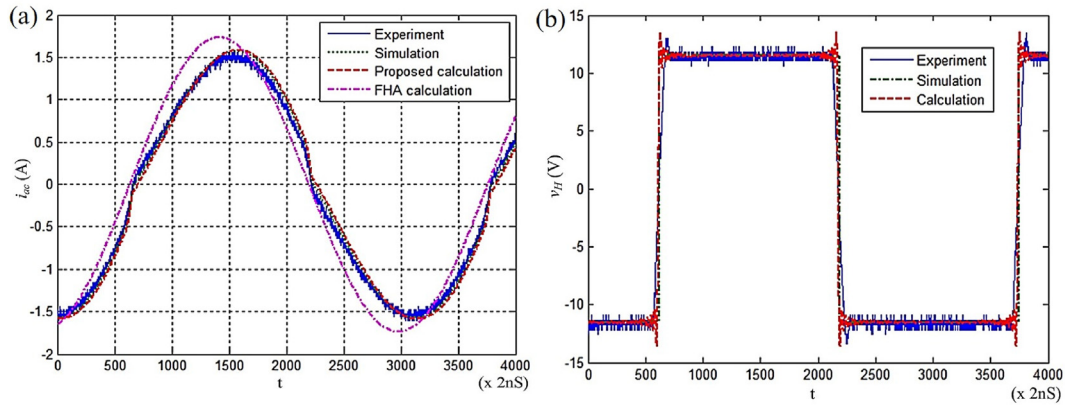


Fig. 10. (a) i_{ac} and (b) v_H primary switching waveforms from simulation, calculation and experiment for SS topology ($N = 50$, $V_{in} = 11.52$ V $R_o = 10 \Omega$).

sharp transient waveform. Therefore, the deadtime is not included in the simulation and the calculation for this case. Fig. 10(a) compares the results of the simulation, calculation and experiments, where satisfactory agreements are obtained for the proposed method. The proposed method has high agreement with the time-domain simulation, whom has the same parameters. On the other hand, as the inductances of the coils are measured by LCR meter with only 500 mV excitation, the non-linear behavior at high current brings error to the lump parameters. The peak current in the experimental results is approximately 6% lower than the counterparts. The zero crossing for the theoretical calculation has slightly bigger error, while the simulation matches better with the experiment. The peak value of FHA is 16% higher than that of the experiment. The output power of all cases is listed in Table 3. The FHA has 18% error, while the proposed method has less than 1% error. For SS topology, the proposed method exhibits the very detail

of steady state resonant waveforms.

Fig. 11 shows the same set of comparison objects in LCC topology. To output the same power as SS topology, V_{in} is boosted to 17 V. The primary side current is not large enough to realize ZVS, resulting in a slow slope of the switching transient during dead-time. When aligning the phase of current waveforms from different methods, a small phase shift from experimental result is observed in Fig. 11(b). This phase shift is caused by the impedance mismatch between C_{fp} branch and C_p branch. More precisely, the ESR on C_p branch, namely the core loss, increases with high order harmonic and high current condition. This effect become noticeable in LCC topology, while SS topology forces the current to the same phase automatically. In fact, the core loss is non-linear for different frequency components, which breaks the assumption that the resonant network is linear. Both the simulation tool and the calculation do not consider the non-linear core loss and they match with each

Table 3

Output power comparison ($N = 50$).

		FHA	Simulation	Proposed	Experiment
SS	P_o (W)	12.14	10.34	10.22	10.26
	Percentage	118.3%	100.7%	99.6%	100%
	P_o (W)	5.91	5.12	5.07	5.13
	Percentage	115.2%	99.8%	98.8%	100%
LCC	P_o (W)	12.12	11.47	11.13	10.07
	Percentage	120.4%	113.9%	110.6%	100%
	P_o (W)	6.22	5.62	5.27	5.12
	Percentage	121.5%	109.8%	102.9%	100%

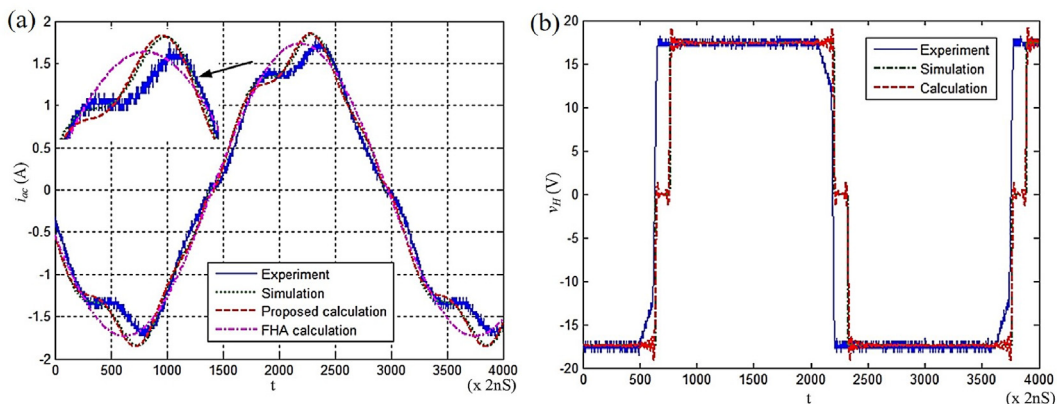


Fig. 11. (a) i_{ac} and (b) v_H primary switching waveforms from simulation, calculation and experiment for LCC topology ($N = 50$, $V_{in} = 17.4$ V $R_o = 10 \Omega$).

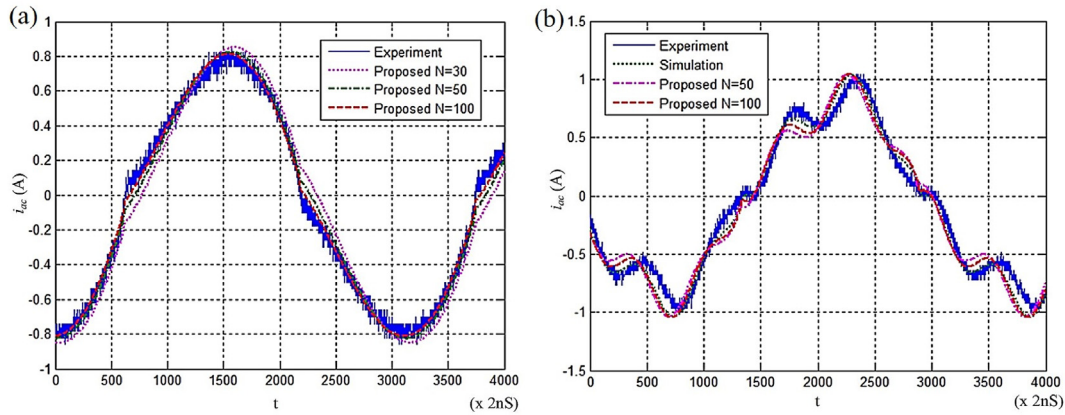


Fig. 12. (a) i_{ac} waveforms from proposed calculation ($N = 30, 50$ and 100) and experiment at lighter load condition (SS, $V_{in} = 11.1$ V, $R_o = 20$ Ω). (b) i_{ac} waveforms from proposed calculation ($N = 50$ and 100) and experiment at lighter load condition (LCC, $V_{in} = 9.4$ V, $R_o = 20$ Ω).

other. As shown in Table 3, this effect also results in a low output voltage than expected in simulation and calculation. Fig. 11(a) draws a similar conclusion as Fig. 10(a), where the proposed method has high agreement with the time-domain simulation results. The experimental waveform shows bigger mismatch around the peak current. The FHA calculation waveform has a larger envelope; thus, a bigger output power error exists among these methods.

The output power comparison of the two topologies is listed in Table 3. The heavy load conditions are corresponded to Figs. 10 and 11. The half load conditions are corresponded to Fig. 12. The proposed method shows high accuracy for SS topology within 2% error. The error become larger in LCC topology as the lump parameter extraction does not fully describe the transformer's behavior. From LCR meter, the AC resistor of the coil is 5 times larger at the third

harmonic than the fundamental frequency. The second factor is the non-linear behavior of inductor over current. Despite this parameter variation, the proposed method shows good agreement with the time-domain simulation results, except for the 5 W LCC case in DCM as shown in Fig. 12(b). The FHA method, as the initial step of the proposed method, has relatively larger error in all cases above. Fig. 12(a) give an example of a lighter load condition for SS topology. The reconstructed time-domain results with different N are plotted. As N increases, the accuracy increases at the cost of calculation resources. Therefore, it becomes a practical limitation when the converter works near to the boundary mode even it is in CCM.

Fig. 12(b) gives another example of lighter load condition for LCC topology. It can be seen that the error function increases slowly as approaching DCM region. However, there is no clear cut of the error function around the boundary, or sudden failure of the proposed

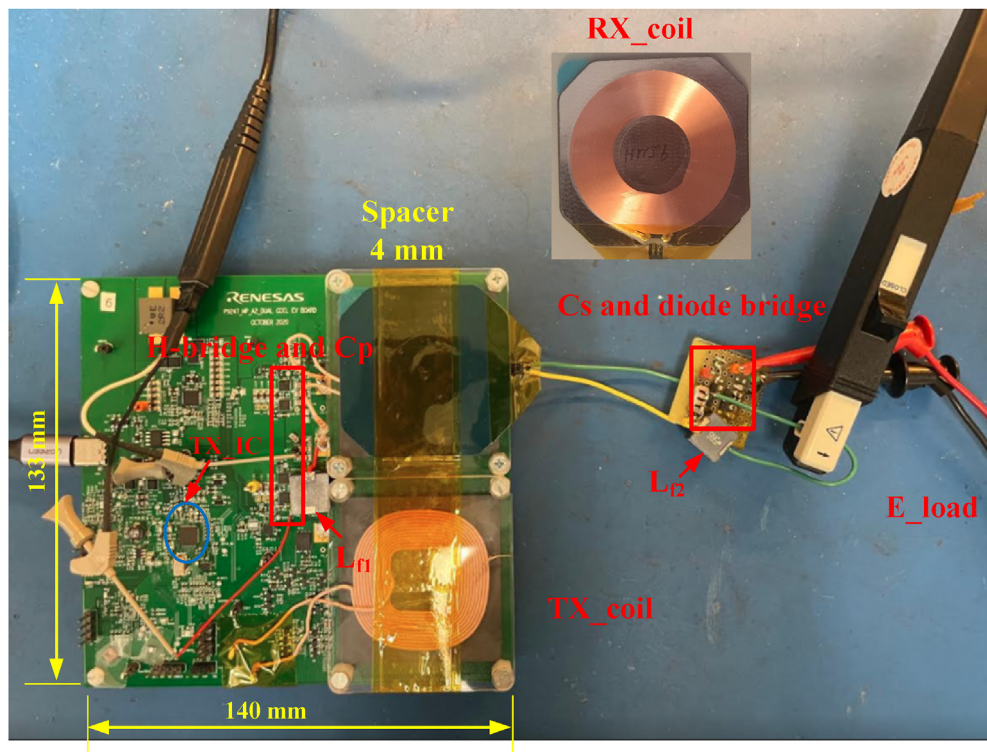


Fig. 13. Photo of the prototype.

method. A light DCM case shows that the proposed method can still reveal some details in DCM region, which indicates the proposed method has potential to be extended to DCM region in future study. In summary of the comparison:

- 1) The proposed method has an equivalent performance as the time-domain simulation in terms of the accuracy. The conventional FHA method could only show the fundamental component with higher error than the proposed method.
- 2) The proposed method has benefit on computational cost over the time-domain simulation when the dimension of the matrix is relatively low. The conventional FHA method is the fastest way for initial estimation.
- 3) Both the proposed method and conventional FHA method are limited to CCM case as the assumption of switching function. The time-domain simulation does not have such limitation.

4. Conclusions and future work

In this study, the first order frequency-domain analytical model for resonant converters in CCM is developed for WPT applications. The proposed model is a general model for resonant converter that allowed arbitrary input and output configurations, which has high flexibility over existing frequency-domain models. Comparing to the traditional FHA method, the proposed model has 10% improvement on the estimation of output power and shows the very details of steady state waveforms. It is an effective tool to guide the WPT design at optimization stage. The proposed analytical model is a complete model for the whole resonant converter. Due to the limitations of the modeling of diode-bridge rectifier in other existing studies, it is hard to combine the resonant network into the model without simplification. The proposed model is the first of its kind that contains all the information of the non-linear diode-bridge rectifier. The analytical model consists of only one variable to achieve simple and fast searching algorithm for the steady state. The proposed model has high potential to enable auto calculation of the resonant converter's steady state parameters. The parabolic interpolation algorithm is chosen to demonstrate the efficacy of the analytical model. From the computational complexity analysis, such frequency-domain model has benefit over time-domain model when the rank of the matrix is not too high. From circuitry point of view, LCC topology has more high order harmonics than SS topology, which needs higher rank of the matrix to reveal the detail waveforms. Therefore, a more complex topology would expect more computational complexity on the proposed model.

One limitation for the presented model is that N tends to be high if the topology has rich harmonics in high orders, which required new searching algorithm or optimized matrix operation in future study. The other limitation is the lack of consideration on the non-linear power loss characteristic in the resonant network. When the major part of error from the diode-bridge rectifier is correctly modeled, the non-linear power loss becomes dominant and affects the accuracy as the experiment shows. For future work, similar approach can be expanded to DCM operation. Nevertheless, the shape of the switching function needs to be pre-determined according to a specific case, which results in the difficulty of finding a general expression for DCM.

Credit author statement

Hulong Zeng: Conceptualization, Resources, Writing – original draft. **Pengcheng Jiao:** Resources, Supervision, Writing – original draft. **Yonghui Liu:** Investigation, Data curation. **Ujjwal Karki:** Investigation, Data curation. **Fang Z. Peng:** Conceptualization, Supervision, Review & Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This study is supported in part by the Key Research and Development Plan of Zhejiang, China (2021C03180), and the Key-Area Research and Development Program of Guangdong Province (2021B0707030002). PJ acknowledges the Startup Fund of the Hundred Talent Program at the Zhejiang University, China.

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